

Harry Nguyen

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EDUCATION

Georgia State University

Expected May 2027

Bachelor of Science in Computer Science • **GPA: 3.75/4.0**

- Coursework: Distributed Systems, Operating Systems, Database Systems, Computer Networks, Data Structures
- Honors: Presidential Scholarship, Anita Andrew Fellowship, Roxie Alexander Memorial Award
- Activities: Amazon University Event – BEN Tech Cloud Connect ATL

EXPERIENCE

Google | Software Engineer Intern — Chrome Browser

May 2025 – Aug 2025

- Designed **C++ IPC transport layer** with Protocol Buffers — analyzed serialization tradeoffs (schema evolution, cross-language compatibility); shipped to Chrome stable channel serving **3B+ active users** at **sub-50ms p99**; adopted by senior Chrome engineers at **95% test coverage**
- Profiled settings navigation bottleneck at p99 1,200ms; designed and implemented **lock-free concurrent trie search** using CAS (compare-and-swap) to eliminate mutex contention — **96% latency reduction**, zero production regressions
- Architected event-driven TypeScript/React system with observer pattern decoupling UI state propagation — accelerated feature delivery **68%** across **25K+ lines** of Chromium at **95% test coverage**
- Produced design documents and code reviews with Chrome infrastructure team; changes adopted into production branch

TiMoto AI | Software Engineer

Sep 2025 – Present

- Architected multi-AZ ECS Fargate with **Terraform IaC**, CloudWatch observability, and circuit breaker fault-isolation pattern — **99.9% uptime**, **44% cost reduction** (\$40–60/month); auto-rollback on health check failure; primary engineer for backend, cloud infra, and ML serving on 3-person team
- Designed **gRPC inter-service layer with exactly-once semantics**; resolved production deadlock by tracing shared resource acquisition conflicts and redesigning call sequencing — **100% evaluation success rate** at **sub-50ms p99**
- Architected **vLLM inference engine with PagedAttention** — eliminated KV cache memory fragmentation under concurrent load; zero OOM failures at production traffic
- Maintained distributed production systems via root-cause analysis, runbooks, and post-mortems; on-call rotation across gRPC, PostgreSQL, and ML serving layers

Develop for Good | Software Engineer Intern

May 2024 – Aug 2024

- Designed stateless BaaS on AWS — JWT auth for horizontal scalability; auto-scaling supporting **500+ concurrent users**, **90% deployment time reduction** via CI/CD
- Diagnosed N+1 query bottleneck at 3s+; redesigned with PostgreSQL indexing achieving **sub-100ms** for 10,000+ records

PROJECTS

Pulumi – Contributor | Go, TypeScript, Infrastructure-as-Code (IaC)

github.com/pulumi

- Submitted Go CLI features and bug fixes enabling multi-cloud (AWS/Azure/GCP) provisioning — under active review by core maintainers
- Analyzed **Raft/Paxos consensus** in distributed state synchronization layer — studied linearizability guarantees, correctness under concurrent operations, and fault-tolerance under partial failures; directly applicable to distributed namespace and replication systems

SKILLS

Distributed Systems: gRPC, Protocol Buffers, Circuit breakers, Exactly-once semantics, Fault tolerance, Raft/Paxos, CAS, Multi-AZ, Event-driven architecture

Languages: C++, Go, Python, TypeScript, Java, JavaScript, SQL, Bash, Rust

Cloud & Infrastructure: AWS (ECS Fargate, EKS, EC2, S3, RDS), Terraform, K8s, Docker, CI/CD, CloudWatch, Prometheus, Grafana, GCP

ML & AI Infrastructure: vLLM, PagedAttention, Continuous Batching, LLM-as-a-judge evaluation, LangChain, PyTorch, LoRA/QLoRA

Frameworks & Databases: FastAPI, Django, Node.js, React, REST APIs, PostgreSQL, Redis, MongoDB

AI Dev Tools: Claude Code, GitHub Copilot, Codex, Cursor